Arnav Raj

Roll No.: 20244036
Bachelor of Technology
Electronics and Communications Engineering
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Education

•Motilal Nehru National Institute of Technology, Allahabad *2024-2028*
Bachelor of Technology in Electronics and Communications Engineering CPI: 9.02 (till 2nd sem)

•Amenity Public School, Uttarakhand *2024*
Central Board of Secondary Education Class XII Percentage: 97.8%

•Amenity Public School, Uttarakhand *2022*
Central Board of Secondary Education Class X Percentage: 97.6%

Projects

•MarlOS — Decentralized Multi-Agent Reinforcement Learning OS *November 2025*
Decentralized, auction-based distributed OS using MARL for fairness-aware, self-regulating resource management.

- Tech Stack: Python, ZeroMQ (PUB/SUB), PyTorch, Ed25519
- Engineered a P2P distributed OS that eliminates centralized orchestrators (e.g., Kubernetes) by enabling autonomous, cryptographically authenticated nodes.
- Implemented a multi-agent reinforcement learning (MARL) core for cooperative, auction-based bidding and self-healing behaviors.
- Designed a Fairness-Aware Economic Layer using adaptive tokenomics (MarlCredits) to ensure equitable resource allocation and prevent monopolies.
- Source Code: arnavraj-7/MarlOS

•ICMT: I Can Manage Things *October 2025*
Cross-platform real-time team collaboration and task management mobile app with AI-powered productivity insights

- Tech Stack: React Native, Expo, Zustand, Supabase, Socket.io, Express.js, Node.js, Gemini API
- Developed a real-time team collaboration mobile app enabling users to create and join teams, assign roles, manage tasks, set priorities and deadlines, and attach documents or resource links.
- Integrated Socket.io for instant updates on team invitations, task assignments, and completion notifications.
- Implemented a personalized AI assistant that summarizes ongoing projects, highlights pending tasks, and suggests optimal work schedules.
- Designed hierarchical task structures with subtasks, threaded comments, and resource attachments to enhance collaboration.
- Planned future integrations with Google Calendar and Google Meet for seamless event scheduling and video conferencing.
- Source Code: arnavraj-7/ICMT—I-Can-Manage-Things

•CrowdSpark - Decentralized Crowdfunding Platform *June 2025*
Full-stack Web3 dApp for transparent campaign creation, funding and blockchain integration

- Tech Stack: Next.js, Tailwind CSS, Solidity, Ethers.js, Hardhat, IPFS, Zustand
- Built smart contracts and backend logic for campaign creation, donation tracking, and goal management using blockchain.
- Integrated responsive UI with smooth animations and state management via Zustand.
- Connected front-end and smart contracts via Ethers.js with secure transactions handled on testnet.
- Live Demo: crowd-spark-ten.vercel.app
- Source Code: arnavraj-7/Crowd-Spark

Positions of Responsibility

•SMP Mentor, Motilal Nehru National Institute of Technology Allahabad *Aug 2025 - Present*

Achievements

•- Hackathon (Project: ICMT) October 2025

Technical Skills and Interests

Languages: Java, Python, JavaScript, TypeScript, Solidity
Frontend: React, React Native, Expo, Next.js, HTML, CSS, Tailwind CSS
Backend: Express.js, Node.js, MongoDB
Tools & Technologies: Git, GitHub, Vercel, Render, Ethers.js, Hardhat, MetaMask, IPFS
AI: LangChain
Areas of Interest: Full Stack Development, Mobile App Development, AI Agents, Blockchain & Web3